k^2-5 UNIT SQUARES NEED A SQUARE OF SIDE k FOR EVERY $k \geq 7$

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ABSTRACT. Let $s(n)$ be the side of the smallest square into which n unit squares with pairwise disjoint interiors can be packed, rotations allowed. We prove $s(k^2-5)=k$ for every $k \geq 7$. The cases $k=7,8,9$ are known certificates (ours for $s(44)=7$, Hosono's for $s(59)=8$, Greff's for $s(76)=9$). For all $k \geq 10$ at once we give one family of measures μ_k on $[0,k]^2$: a fixed corner module in each corner, a wall pattern of depth 4 that repeats with period 1 along each side, and Lebesgue measure in the middle. Its total mass is k^2-4D with $D=1340566927759/10^{12}>5/4$, and a locality argument reduces the covering property for every $k \geq 10$ to a single 11×11 board. That board is checked by Daniel's interval checker `zmx2` everywhere except near *tight tiles*: axis-parallel unit squares next to a wall whose mass is exactly 1, which no box-subdivision checker can close. For those we prove a Tight-Tile Lemma by an exact blow-up argument: in coordinates scaled by the tilt, the mass is bounded below on each cell by a concave function built from exact limit values and interval-certified derivative bounds, and this bound is checked at the cell's vertices in rational arithmetic.

1. INTRODUCTION

Write $s(n)$ for the side of the smallest square containing n unit squares with pairwise disjoint interiors, each free to rotate. Since $s(k^2)=k$ and s is nondecreasing, $s(n) \leq k$ for $(k-1)^2 < n \leq k^2$, and one asks how many squares can be removed from the $k \times k$ grid before a smaller container suffices. Friedman's survey [4] records the history. Nagamochi [10] studied $s(k^2-1)$ and $s(k^2-2)$; Karakuş [8] later gave a complete proof of $s(k^2-1)=k$, filling a gap in the argument for Nagamochi's Lemma 1. The family $s(k^2-3)=k$ is due to Kearney and Shiu for $k=3$ [9], Bentz for $4 \leq k \leq 7$ [1, 2], and Daniel for all $k \geq 6$ [3]; since s is nondecreasing, it also gives $s(k^2-2)=k$ for $k \geq 3$. Daniel also proved $s(k^2-4)=k$ for all $k \geq 5$ by computer-assisted weighted covers (the case $k=9$ was first proved by Hosono; see the record in [3]). Ryu's preprint [11] shows that for every fixed integer $a \geq 1$, $s(k^2-a)=k$ for all sufficiently large k; its explicit bound reaches $a=5$ only for $\log k \geq 182$, so it does not settle any case covered here.

Theorem 1. *$s(k^2-5)=k$ for every integer $k \geq 7$.*

The upper bound $s(k^2-5) \leq k$ is the $k \times k$ grid with five cells empty. The lower bound comes from four sources:

k	$n=k^2-5$	proof of $s(n) \geq k$	record
7	44	our weighted cover [12]	T-252 of [13]
8	59	Hosono's weighted cover [7]	T-066 of [13]
9	76	Greff's weighted cover [5]	T-184 of [13]
≥ 10	$95, 116, \ldots$	the family of this paper	—

The case $k=10$, $s(95)=10$, was proved earlier by Greff with a single certificate (together with $s(96)=10$ [6], T-191 of [13]); our family gives an independent proof, so the argument does not depend on it. The bound $k \geq 7$ cannot be lowered to all k: for $k=3$ and $k=4$, removing five squares does free space, since $s(4)=2$ and $s(11)<3.88$ [4]. The cases $k=5,6$ ($n=20,31$)

Date: Draft, November 13, 2026.

remain open; covers of the kind used here very likely cannot reach them (Remark 18). Since s is nondecreasing, Theorem 1 also gives $s(k^2 - a) = k$ for $a = 1, \dots, 4$ and every $k \geq 7$; those cases were already known.

What is new. Daniel's families for $k^2 - 3$ and $k^2 - 4$ already have the corner-plus-periodic-wall form, and our search reuses his linear program and his checkers. Two things are new. First, the parameters (corner size 4, wall depth 4, element pitch 1/10, a Lebesgue interior) push the saving per corner from about 1.25, the limit Daniel found at pitch 1/5, to $D \approx 1.3406 > 5/4$. Second, this margin forces the cover to be exactly tight on many wall tiles, where the existing box-subdivision checkers stall; Section 6 gives a rigorous argument that closes them.

The idea in one paragraph. A measure on $[0, k]^2$ that gives every unit square mass at least 1 shows that at most its total mass of pairwise disjoint unit squares fit (Section 2). Lebesgue measure has total k^2; we lower this by more than 5 by working near the walls and corners. Our measure μ_k is Lebesgue measure in the middle, a periodic pattern of weighted segments along each wall, and a fixed pattern in each corner (Figure 2). Each corner saves $D \approx 1.3406$, so the total is $k^2 - 4D < k^2 - 5$. Because the pattern is periodic, checking that every unit square gets mass at least 1 for every $k \geq 10$ reduces to one board, $k = 11$. Most of that board is checked by an existing interval checker; the hard part is a family of squares next to the walls that get mass exactly 1 in the limit of zero tilt, for which we prove a separate lemma.

Structure of the proof. Section 2 recalls the standard reduction from packings to weighted covers. Section 3 defines the family μ_k and computes its mass, and Section 4 shows that validity of the single measure μ_{11} implies the lower bound for every $k \geq 10$. Section 5 proves that μ_{11} is valid (Theorem 10), using the Tight-Tile Lemma (Theorem 11), which is proved in Section 6. The proof of Theorem 1 is assembled at the end of Section 5; Figure 1 shows how the pieces fit together. Sections 7–9 describe controls, the search, and remarks.

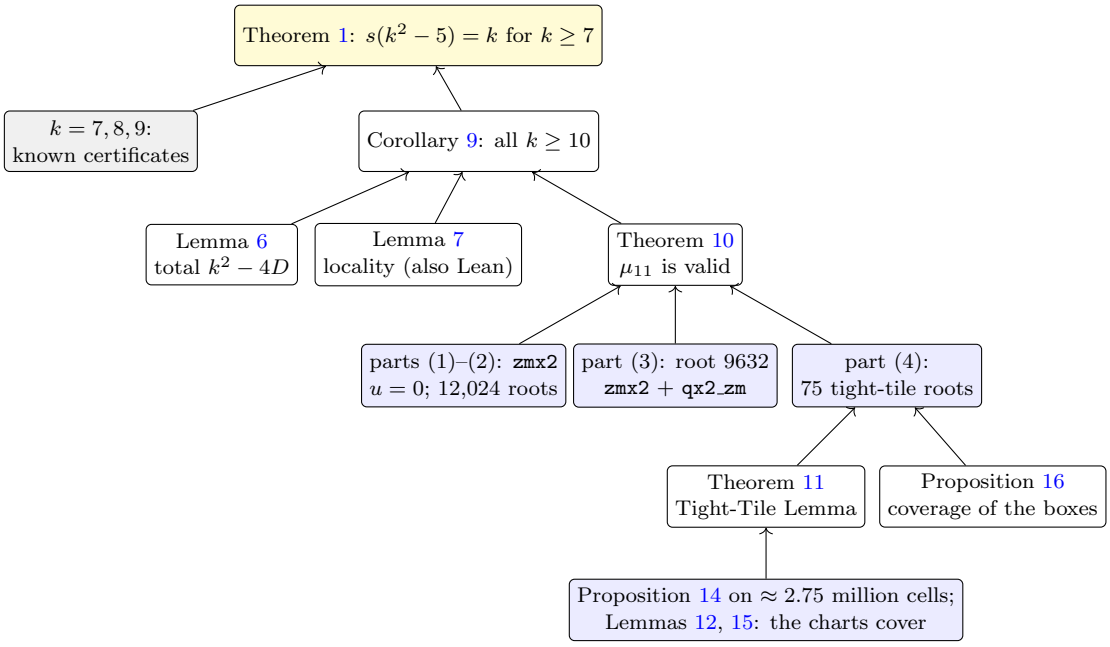


FIGURE 1. Structure of the proof. Shaded boxes are computations; the others are proved in the text.

2. Preliminaries

Notation. For a point $c \in \mathbb{R}^2$, an angle θ and $L > 0$, let $S(c, \theta, L)$ be the closed square of side L centered at c whose sides make angle θ with the axes. A *pose* in $[0, k]^2$ is a closed unit square $P = S(c, \theta, 1) \subseteq [0, k]^2$. Because of the symmetries of the square, θ matters only modulo $\pi/2$; we write $u = \tan(\theta/2)$, so that $\theta \in (-\pi/4, \pi/4]$ corresponds to $u \in (-\tan\frac{\pi}{8}, \tan\frac{\pi}{8}]$. All measures are finite Borel measures on a square $[0, k]^2$; the *mass* of a pose P is $\mu(P)$.

From packings to covers. We use the two standard lemmas in the closed form of Daniel [3], where both are proved in Lean 4 (`packing_le_measure`, `not_packs_of_measure`); short proofs are also given in our $s(44)$ paper [12].

Lemma 2 (scaling)**.** *If n unit squares with pairwise disjoint interiors lie in a square of side $L < k$, then there are n pairwise disjoint poses in $[0, k]^2$.*

Lemma 3 (weighted counting)**.** *Let μ be a finite Borel measure on $[0, k]^2$ with $\mu(P) \geq 1$ for every pose P. Then at most $\mu([0, k]^2)$ poses are pairwise disjoint.*

Together: if such a μ has $\mu([0, k]^2) < n$, then n unit squares do not fit in any square of side $L < k$ (Lemma 2 would give n disjoint poses, contradicting Lemma 3), so $s(n) \geq k$.

Definition 4. A measure μ on $[0, k]^2$ is *valid* if $\mu(P) \geq 1$ for every pose P in $[0, k]^2$.

All our measures are finite sums of two kinds of pieces: an axis-parallel segment γ carrying a weight $w > 0$ spread uniformly by length (so γ itself has mass w), and Lebesgue measure on an axis-parallel square.

3. The family of measures

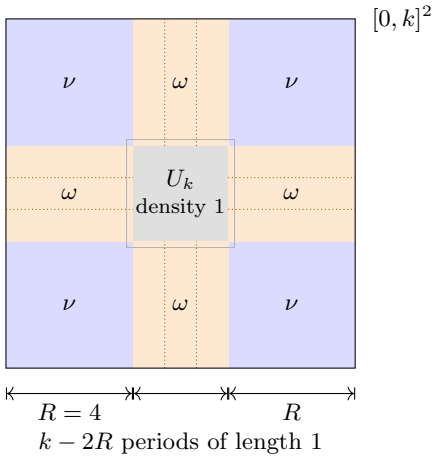


FIGURE 2. Layout of μ_k (drawn for $k = 11$). The corner module ν fills each $R \times R$ corner; the wall pattern ω, of period 1 along the wall (dotted lines), fills the four bands between the corners; Lebesgue measure (grey) covers $U_k = [\frac{19}{5}, k - \frac{19}{5}]^2$, which overlaps the inner edge of the bands. Increasing k by one adds one period to each band and leaves the corners unchanged.

Fix $R = 4$. All coordinates below are multiples of $1/10$, and all weights are exact rationals.

Definition 5. The family is determined by two finite sums of weighted axis-parallel segments, all with endpoints in $\frac{1}{10}\mathbb{Z}^2$:

(1) a *corner module* ν in $[0,R]^2$, invariant under $(x,y)\mapsto(y,x)$;
(2) a *wall period* ω_0 in $[0,1]\times[0,R]$ (first coordinate along the wall, second the distance from the wall), whose segments start at a first coordinate in $[0,1)$. Its periodic extension $\omega=\sum_{m\in\mathbb{Z}}\big(\omega_0+(m,0)\big)$ is invariant under $(x,y)\mapsto(-x,y)$.

For an integer $k\geq 2R$, the measure μ_k on $[0,k]^2$ is the sum of the following pieces:

- four *corner pieces*: ν in the corner $[0,R]^2$, and its images in the other three corners under the symmetries of $[0,k]^2$;
- four *wall pieces*: the bottom wall piece consists of the segments of ω that lie in the strip $R\leq x\leq k-R$; the other three are its images under the symmetries of $[0,k]^2$;
- Lebesgue measure on the square $U_k=[\frac{19}{5},\,k-\frac{19}{5}]^2$.

Since every segment has its endpoints on the grid $\frac{1}{10}\mathbb{Z}^2$ and the lines $x=R$, $x=k-R$ are grid lines, no segment crosses them, so "the segments that lie in the strip" is unambiguous. By construction, and because ω is symmetric, μ_k is invariant under the eight symmetries of $[0,k]^2$; for $k=10,\dots,13$ and $k=20$ this was also checked in exact arithmetic on the built measures. (The family file records, in its header, the part of U_k that lies in the wall and corner strips as belonging to those pieces; this is bookkeeping and gives the same μ_k.)

The exact data (2014 rational variables) are in the file `sol_exact_family.txt` [15]. The measure μ_{11} is the file `sol_exact_box11.txt` (SHA-256 `d259fed5...12099`): 13,244 weighted segments with coordinate denominator 10 and mass denominator $2\cdot10^{12}$, plus Lebesgue measure on $U_{11}=[19/5,36/5]^2$.

Figure 3 shows the actual measure μ_{11}.

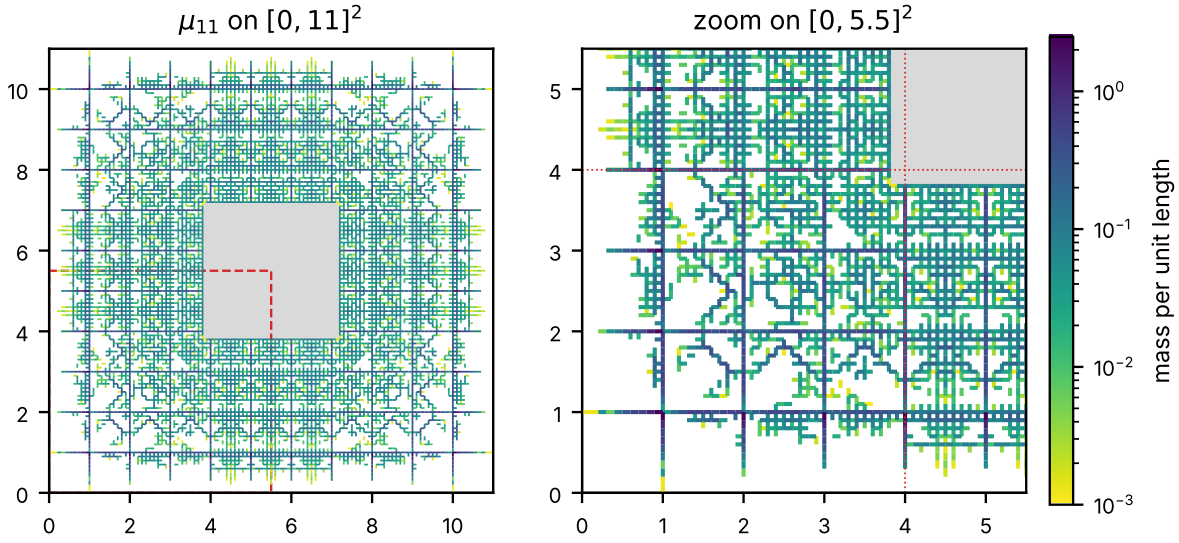


FIGURE 3. The measure μ_{11}: every segment drawn in a colour given by its mass per unit length (segments lighter than 10^{-3} per unit length are omitted); grey is Lebesgue measure on U_{11}. Right: the corner $[0,5.5]^2$; the dotted red lines are $x=R$ and $y=R$, where the corner module meets the wall bands.

Lemma 6 (mass)**.** *For every integer $k\geq 2R$,*

$$\mu_k([0,k]^2)=k^2-4D,\qquad D=\frac{1340566927759}{10^{12}}\approx 1.3405669.$$

In particular $\mu_k([0,k]^2) < k^2 - 5$ *for every such* k*, since* $D > 5/4$.

Proof. On each side, the segments of ω in the strip $R \leq x \leq k-R$ are the translates $\omega_0 + (m, 0)$, $m = R, \ldots, k-R-1$, together with the segments of ω_0 on the line $x = 0$ translated to $x = k-R$. So the total mass is

$$4\,\nu([0,R]^2) + 4(k-2R)\,\omega_0([0,1]\times[0,R]) + 4\,\omega_0(\{0\}\times[0,R]) + \Big(k - \frac{38}{5}\Big)^2,$$

a quadratic polynomial in k with leading coefficient 1. The data are normalized so that it equals $k^2 - 4D$; this is checked in exact rational arithmetic from the family file, and independently by building μ_k for $k = 10, 11, 12, 13, 20$ and summing all weights. For $k = 11$ the total is $28909433072241/250000000000 = 121 - 4D$. □

Thus D is the saving per corner: away from the corners the measure is as economical as the grid itself.

4. Reduction to one board

The measure μ_k looks the same, near every point, as μ_{11} near some point: near a wall it repeats with period 1, and near a corner it does not depend on k. Since a unit square only sees a window of side $\sqrt{2}$, its mass under μ_k equals the mass of a translated square under μ_{11} (Figure 4).

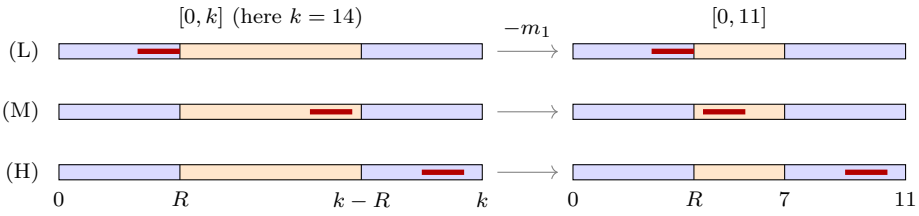


FIGURE 4. The three cases of Lemma 7 in one coordinate. The red bar is the shadow $[a, b]$ of the unit square (length at most $\sqrt{2}$). Blue: within R of a side; orange: the periodic band. Near the left side (L) nothing moves; in the band (M) an integer shift brings the shadow into $(R, R+1+\sqrt{2}]$; near the right side (H) the shadow moves with the side.

Lemma 7 (locality)**.** *Let* $k \geq 10$*. For every pose* P *in* $[0,k]^2$ *there is* $(m_1, m_2) \in \mathbb{Z}^2$ *such that* $P' = P - (m_1, m_2)$ *is a pose in* $[0,11]^2$ *and* $\mu_k(P) = \mu_{11}(P')$.

Proof. Let $[a_1, b_1] \times [a_2, b_2]$ be the smallest axis-parallel box containing P. Its sides have length $|\cos\theta| + |\sin\theta| \leq \sqrt{2}$, and $\mu_k(P)$ depends only on the restriction of μ_k to this box. We choose m_1 from $[a, b] = [a_1, b_1]$ (and m_2 from $[a_2, b_2]$ in the same way) by three cases:

(L) $a \leq R$: take $m_1 = 0$. Then $b \leq R + \sqrt{2} < 5.42$.

(M) $R < a$ and $b < k - R$: take $m_1 = \lceil a \rceil - R - 1$, so that $a - m_1 \in (R, R+1]$ and $[a, b] - m_1 \subseteq (R,\ R+1+\sqrt{2}] \subseteq (R,\ 7) = (R,\ 11-R)$.

(H) $b \geq k - R$: take $m_1 = k - 11$. Then $a \geq k - R - \sqrt{2} > R$ (as $k \geq 10$), so case (L) does not apply, and $[a, b] - m_1 \subseteq [11 - R - \sqrt{2},\ 11]$.

The cases are exhaustive, and (L) and (H) are exclusive because $b - a \leq \sqrt{2} < k - 2R$.

In each case, every piece of μ_k that meets the strip $[a, b] \times \mathbb{R}$ coincides, after translation by m_1, with the corresponding piece of μ_{11} on $([a, b] - m_1) \times \mathbb{R}$, as far as its dependence on x is concerned:

- In case (L) the strip lies in $\{x < 5.42\}$. Pieces attached to the right side of $[0,k]^2$ (right corner pieces, right wall piece) lie in $\{x \geq k - R\} \subseteq \{x \geq 6\}$, and U_k extends to $x = k - \frac{19}{5} \geq 6.2$; so in this strip μ_k consists of the left pieces, the segments of the bottom and top wall pieces with $x \leq 5.42$ (those of ω in $R \leq x \leq 5.42$, the same for every $k \geq 10$ since $k - R \geq 6$), and U_k near its left edge. All of these are defined relative to the left side $x = 0$ in the same way for every $k \geq 10$, and in particular for μ_{11}.
- In case (M) the strip lies in the open strip $R < x < k - R$, which no corner piece and no left or right wall piece meets (those lie in $x \leq R$ or $x \geq k - R$). There the bottom and top wall pieces coincide with ω (and its image), because a segment of ω that meets the open strip lies in $R \leq x \leq k - R$; ω is invariant under integer translations in x; and U_k is invariant under horizontal translation within $\frac{19}{5} \leq x \leq k - \frac{19}{5}$, which contains the strip. The integer translation by m_1 maps the strip into $R < x < 7$, where μ_{11} has the same description.
- Case (H) is case (L) reflected in $x \mapsto k - x$ (respectively $x \mapsto 11 - x$), by the symmetry of μ_k and μ_{11}: the strip lies within distance $R + \sqrt{2}$ of the right side, where both measures are described relative to that side in the same way.

The same holds in y with m_2. Each piece of μ_k is attached either to one side or corner, or is periodic, or is U_k, independently in x and in y: a corner piece is attached to two sides, and it meets the box only when the cases in x and in y are both (L) or (H) for those sides, with zero shift relative to them; a wall piece is attached to one side in one coordinate and periodic in the other; U_k is a product of two intervals. Therefore the restriction of μ_k to $[a_1, b_1] \times [a_2, b_2]$ is the translate by (m_1, m_2) of the restriction of μ_{11} to $([a_1, b_1] - m_1) \times ([a_2, b_2] - m_2) \subseteq [0, 11]^2$. Hence $P' = P - (m_1, m_2)$ lies in $[0, 11]^2$ and $\mu_{11}(P') = \mu_k(P)$. □

Remark 8 (formal proof). The argument follows Daniel's reduction for $k^2 - 3$ and $k^2 - 4$ (file `BentzFam.lean` in [3]). We ported it to the 1/10 grid in Lean 4 (files `Fam10.lean` and `Grid10.lean`); the instance proves

$$\texttt{Valid11} \implies \forall k \geq 10,\ s(k^2 - 5) = k$$

(theorem `k2m5_of_valid11`), where `Valid11` states that μ_{11}, read verbatim from the board file, is valid. It uses only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The kernel also checks that the board file is μ_{11} of the family and that the total mass of μ_k is $k^2 - 4D$ for every $k \geq 9$; this was built for the final measure. As an independent check, a script written from scratch confirms on the 1/10 grid that every 1.6×1.6 window of μ_k is an integer translate of a window of μ_{11} for $k = 10, 12, 13, 20$, and it detects a one-weight mutation of μ_{12} (176 windows fail).

Corollary 9. *If μ_{11} is valid on $[0, 11]^2$, then $s(k^2 - 5) \geq k$ for every $k \geq 10$.*

Proof. By Lemma 7, μ_k is valid on $[0, k]^2$ for every $k \geq 10$. By Lemma 6 its total mass is $k^2 - 4D < k^2 - 5$, so Lemmas 2 and 3 give $s(k^2 - 5) \geq k$. □

5. Validity of the 11×11 board

Theorem 10. *The measure μ_{11} is valid on $[0, 11]^2$: every pose P in $[0, 11]^2$ has $\mu_{11}(P) \geq 1$.*

The proof is computer-assisted. Pose space is three-dimensional (the center and the angle). The checker `zmx2` [3] splits it into *root boxes* (a grid of initial boxes in center and u), subdivides each root box adaptively, and certifies each sub-box by a rigorous lower bound on the mass of all poses in it. It computes the geometry in binary64 interval arithmetic with outward rounding and compares masses as exact integers. In *first-order* mode it also uses the first-order behavior in u near $u = 0$. A root box is *certified* when all of its sub-boxes are, up to a depth cap.

Proof of Theorem 10. The measure μ_{11} is invariant under the eight symmetries of the square (checked in exact arithmetic), and these symmetries act on poses. So it suffices to treat poses with center in $[0, \frac{11}{2}]^2$ and $u \in [0, \tan\frac{\pi}{8}]$. We split this set into four parts.

(1) **Untilted poses, $u = 0$.** The mass of an axis-parallel pose is a piecewise-affine function of its center whose minimum is attained at finitely many limit points; Daniel's "Lemma Z" [3] reduces the check to these points. Three checks agree that every untilted pose has mass ≥ 1: Lemma Z at 10,000 one-sided limit corners in exact arithmetic (minimum exactly 1, attained at 750 of them); an independent exact evaluation at 7,717 corners (minimum exactly 1); and `zmx2` in untilted mode on all 2,500 root boxes (all certified).

(2) **Tilted poses, general position.** `zmx2` in first-order mode with depth cap 40 certifies 12,024 of the 12,100 root boxes. 76 root boxes remain.

(3) **Root box 9632.** One remaining root box lies at a corner of U_{11}, away from the walls. Its uncertified sub-boxes are certified by Daniel's exact-rational checker `qx2_zm.py`, applied to the restriction of μ_{11} to the 3/4-neighborhood of the root box together with its images under the symmetries. Concretely, `zmx2` leaves 6,850 sub-boxes there, all inside the box $\Omega = [4.3224, 4.3264] \times [4.3199, 4.3239] \times [0.0229, 0.0272]$ in (x, y, u), which lies in a single root box of `qx2_zm.py`. That checker (depth 18, 2,357 boxes) certifies this root box except 83 sub-boxes, none of which meets any of the 6,850; a script checks these containments and disjointness in exact arithmetic. Every pose of the root box is therefore certified by one of the two checkers. (Removing segments can only lower masses, so a certificate for the restricted measure is one for μ_{11}.)

(4) **The other 75 root boxes** are neighborhoods of tight tiles (Section 6). By Proposition 16, every sub-box that `zmx2` leaves uncertified in them is contained, in exact arithmetic, in the union of the set of poses covered by the Tight-Tile Lemma (Theorem 11) and the set of untilted poses covered by part (1).

The four parts cover all poses, so every pose has mass at least 1. □

Proof of Theorem 1. The upper bound $s(k^2 - 5) \leq k$ holds because $k^2 - 5$ unit squares fit in the $k \times k$ grid. For the lower bound, the cases $k = 7, 8, 9$ are the certificates listed in the introduction [12, 7, 5]. For $k \geq 10$, Theorem 10 shows that μ_{11} is valid, and Corollary 9 gives $s(k^2 - 5) \geq k$. □

6. The Tight-Tile Lemma

6.1. **The obstruction.** Take the left wall $x = 0$; the other walls follow by symmetry. In the zone $0 \leq x \leq R$ the measure consists of the left wall piece, which repeats with period 1 in y for $R \leq y \leq 11 - R$, the corner modules near $y = 0$ and $y = 11$, and Lebesgue measure on $\frac{19}{5} \leq x \leq R$. The *tile* at depth $j \in \{1, 2, 3\}$ and height y is the axis-parallel unit square with center (d_0, y), $d_0 = j + \frac{1}{2}$. For the optimal measures of this kind the mass of such tiles, as a one-sided limit at $u = 0$, equals exactly 1 along whole intervals of y: the linear program pushes these constraints down to equality. We call these tiles *tight* (Figure 5).

A sub-box whose closure contains a tight limit pose can never be certified by a bound of the form "mass at a point of the box minus a Lipschitz constant times the box size," however small the box is. Of the 76 root boxes that `zmx2` leaves, 75 are exactly the neighborhoods of tight tiles. The idea of the Tight-Tile Lemma is to blow up these neighborhoods: in coordinates scaled by the tilt, the mass has an exact limit as the tilt tends to 0, and the first-order growth with the tilt can be bounded rigorously.

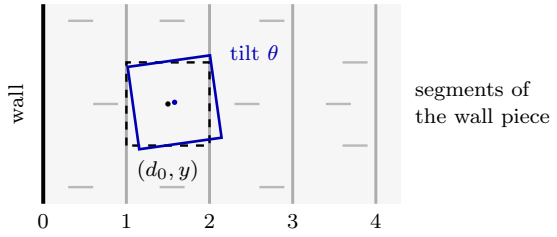


FIGURE 5. A tight tile at depth 1 (dashed) and a nearby pose (blue), with the tilt exaggerated. The tile lies between the lines $x = 1$ and $x = 2$, which carry segments of the wall piece; in the limit of zero tilt its mass is exactly 1. A pose that is shifted by δ and tilted by θ gains or loses pieces of segments near its edges at first order in θ; the lemma shows that the net change is never negative.

6.2. **Statement.** Write the center of a pose as $(d_0 + \delta,\, y)$ and its angle as $\tan(\theta/2) = \sigma u$ with $\sigma \in \{+1, -1\}$ and $u > 0$ (in this section u denotes $|\tan(\theta/2)|$). Let

$$u_0 = \tfrac{1}{4000}, \qquad K = 4, \qquad \delta_{\max} = K u_0 = \tfrac{1}{1000},$$

and define the height ranges

$$\begin{aligned} I(3/2) &= [1.4, 1.7] \cup [2.4, 2.7] \cup [3.3, 5.5], \\ I(5/2) &= [1.4, 1.7] \cup [2.3, 2.7] \cup [3.3, 3.7] \cup [3.8, 5.5], \\ I(7/2) &= [1.4, 1.7] \cup [2.3, 2.7] \cup [3.3, 3.7] \cup [4.1, 4.5]. \end{aligned}$$

Theorem 11 (Tight-Tile Lemma)**.** *Let $d_0 \in \{3/2, 5/2, 7/2\}$, $\sigma \in \{+1, -1\}$ and $y \in I(d_0)$. Every pose with center $(d_0 + \delta, y)$ and $\tan(\theta/2) = \sigma u$, where $0 < u \le u_0$ and $|\delta| \le \delta_{\max}$, has μ_{11}-mass at least 1. The same holds with $u_0 = 1/2000$ and $\delta_{\max} = 1/500$ on five further ranges of y, each extended by $1/500$ at both ends: $[4.2, 4.3]$ and $[4.5, 4.6]$ for $(d_0, \sigma) = (3/2, +1)$; $[3.8, 3.9]$ and $[5.4, 5.5]$ for $(3/2, -1)$; and $[4.4, 4.5]$ for $(5/2, -1)$.*

The proof occupies the rest of this section. It is a computer-assisted case analysis in which every step is either an exact rational computation or a rigorously rounded interval enclosure. The mathematical part of the proof is given in full below: the exact chord formula, the charts and their covering (Lemma 12), the cell certificate (Lemma 13 and Proposition 14), and the corner zones (Lemma 15). The computational part, certifying every cell, is done by the program `ttbox` (about 1,000 lines of Python, in [15]); the technical report [14] documents the implementation, the rounding rules and the logs.

6.3. **Exact chords.** Let $N = 1 + u^2$ and $e_1 = \big((1-u^2)/N,\, 2u/N\big)$, $e_2 = \big(-2u/N,\, (1-u^2)/N\big)$, the unit vectors along the sides of a pose tilted by θ. The pose with center c is $P = \{z \in \mathbb{R}^2 : |(z-c)\cdot e_i| \le \tfrac{1}{2},\ i = 1, 2\}$. Consider a segment $\gamma = \{g + \tau(g' - g) : \tau \in [0, 1]\}$ of weight w. Each of the four half-planes of P reads $\tau A \le B$ with

$$A = \pm\,(g' - g)\cdot N e_i, \qquad B = \tfrac{N}{2} \mp (g - c)\cdot N e_i,$$

for $i = 1, 2$ and both signs. Let $\mathcal{B}^+$ consist of 1 and the ratios B/A with $A > 0$ (upper bounds on τ), and $\mathcal{B}^-$ of 0 and the ratios with $A < 0$ (lower bounds); constraints with $A = 0$ hold either for all τ or for none. Then γ contributes $w \cdot \max(0, \min \mathcal{B}^+ - \max \mathcal{B}^-)$ to $\mu(P)$. The factor $N > 0$ does not change the inequalities, and with it A and B are polynomials in u and c.

6.4. **Blow-up charts.** The key choice is the scale variable: we always use the tilt, $h = u$. Near a base line $y_b \in \frac{1}{10}\mathbb{Z}$, with $t = y - y_b$, the four cases $|\delta| \leq Ku$ or not and $|t| \leq Ku$ or not give four *charts*, each with two *face coordinates* (p, q) and a *slab* condition on h:

chart	poses	face coordinates (p, q)	slab
C_u	$\lvert\delta\rvert, \lvert t\rvert \leq Ku$	$(\delta/u,\ t/u) \in [-K, K]^2$	$0 \leq h \leq u_0$
W_t	$\lvert\delta\rvert \leq Ku < \lvert t\rvert$	$(\delta/u,\ \lvert t\rvert)$	$h \leq \lvert t\rvert/K$
W_d	$\lvert t\rvert \leq Ku < \lvert\delta\rvert$	$(t/u,\ \lvert\delta\rvert)$	$h \leq \lvert\delta\rvert/K$
W_{td}	$Ku < \lvert\delta\rvert$ and $Ku < \lvert t\rvert$	$(\lvert t\rvert, \lvert\delta\rvert)$, two triangles	$h \leq \min(\lvert t\rvert, \lvert\delta\rvert)/K$

Figure 6(a) shows the four charts at a fixed tilt. Between base lines, where $Ku_0 \leq t \leq \frac{1}{10} - Ku_0$, two strip charts are used: Σ_u with face coordinates $(\delta/u,\ y)$ for $|\delta| \leq Ku$ and slab $h \leq u_0$, and Σ_d with face coordinates $(y,\ |\delta|)$ for $|\delta| > Ku$ and slab $h \leq |\delta|/K$.

Lemma 12 (the charts cover)**.** *Every pose of Theorem 11 (with $u_0 = 1/4000$) lies in one of the charts above, with face coordinates in the stated ranges and $h = u$ in the stated slab. The same holds for $u_0 = 1/2000$ on the further ranges.*

Proof. Each set $I(d_0)$ is a finite union of intervals $[y_1, y_2]$ with $y_1, y_2 \in \frac{1}{10}\mathbb{Z}$. Every y in such an interval satisfies either $|y - y_b| \leq Ku_0$ for a base line $y_b \in \frac{1}{10}\mathbb{Z} \cap [y_1, y_2]$, or lies in a strip $[y_b + Ku_0,\ y_b + \frac{1}{10} - Ku_0]$ between two consecutive base lines of the interval. Near a base line, put $t = y - y_b$, so $|t| \leq Ku_0$, and $|\delta| \leq \delta_{\max} = Ku_0$. Exactly one of four cases holds. If $|\delta|, |t| \leq Ku$, the pose is in C_u with $(p, q) = (\delta/u, t/u) \in [-K, K]^2$ and $h = u \leq u_0$. If $|\delta| \leq Ku < |t|$, it is in W_t with $p = \delta/u \in [-K, K]$, $q = |t| \leq Ku_0$ and $h = u < |t|/K$. The case W_d is symmetric. If $Ku < |\delta|$ and $Ku < |t|$, it is in W_{td} with $h < \min(|t|, |\delta|)/K$. In a strip, the same split by $|\delta| \leq Ku$ or $> Ku$ gives Σ_u or Σ_d. For the five further ranges, with $u_0 = 1/2000$, the line charts of the two end lines reach $1/500 = Ku_0$ beyond the range. □

In each chart the center is $c_0 + h\,c_1$, with c_0 affine in (p, q) and c_1 polynomial in (p, q, h), so A and B are polynomials in (p, q, h) with rational coefficients. Common powers of $h > 0$ are divided out exactly. In the limit $h \to 0$ each ratio B/A becomes *affine* in (p, q); this is what makes the tight tiles tractable.

6.5. **The cell bound.** A *cell* is a set $Z = \{(p, q, h) : (p, q) \in \Pi,\ \mathrm{low}(p, q) \leq h \leq \mathrm{high}(p, q)\}$, where Π is a convex polygon with rational vertices and $0 \leq \mathrm{low} \leq \mathrm{high}$ are affine functions. Since Z is convex, its vertices are among the points $(v, \mathrm{low}(v))$ and $(v, \mathrm{high}(v))$ for vertices v of Π. Let $\mathrm{Bx}(Z)$ be the box $\mathrm{bbox}(\Pi) \times [0, \max \mathrm{high}]$.

On a cell, every constraint $\tau A \leq B$ is classified by outward-rounded interval arithmetic on $\mathrm{Bx}(Z)$ as *regular* (A has constant sign, so B/A is C^1), *inactive* ($B \geq \max(0, A)$, so it holds for all $\tau \in [0, 1]$), or *empty* ($B < \min(0, A)$, so it fails for all τ and the chord is 0). On charts of wedge shape, where $h \leq \chi\,\xi$ for a nonnegative face coordinate ξ and a constant $\chi > 0$, the last two tests are made after the exact substitution $h = \eta\,\xi$ with $\eta \in [0, \chi]$ and division by the largest power of ξ; near the apex of a wedge, a second substitution expresses one face coordinate as a bounded multiple ζ of the other. If the auxiliary variables η, ζ make an interval test inconclusive, their ranges are bisected, and the same verdict is required on every piece. For a regular constraint we further require that $A(p, q, 0)$ is a nonzero constant and $B(p, q, 0)$ is affine. A cell on which some constraint fits none of these cases is split.

For each segment, write $b_r^+ = B/A$ for its regular constraints with $A > 0$ together with the constant bound 1, and $b_r^- = B/A$ for those with $A < 0$ together with the constant bound 0 (a constant bound has derivative 0; inactive constraints are dropped, and a segment with an empty constraint contributes 0 on the whole cell). These have affine *limit values* $b_r^{\pm}(0)$ on Π. The

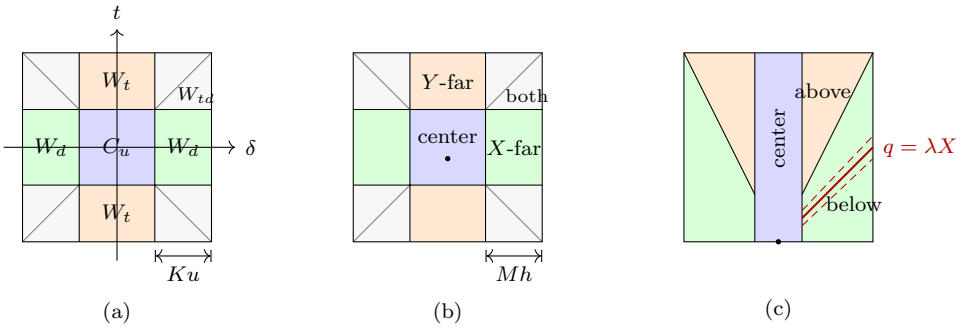


FIGURE 6. The case splits, drawn at a fixed value of the scale $h = u$. (a) The four charts near a base line, in the plane of the offsets (δ, t): C_u where both offsets are at most Ku, W_t and W_d where one of them is larger, W_{td} (two triangles each) where both are. As $u \to 0$ the central square shrinks, and the coordinates δ/u, t/u keep it of fixed size. (b) The same split one level down, around a corner point (p^*, q^*) of a face, with scale Mh. (c) The zones around a corner point on the edge $q = 0$ of a wedge chart, and the near zone (dashed) around a ray $q = \lambda X$, $X = |p - p^*|$.

polygon is split exactly so that, on all of Π, the minimizing upper bound $b^+_{r_*}$, the maximizing lower bound $b^-_{r_*}$ and the sign of the limit chord $f(0) = b^+_{r_*}(0) - b^-_{r_*}(0)$ are fixed. Let

$$\alpha_r \le \inf_{\mathrm{Bx}(Z)} \partial_h b^+_r, \qquad \beta_r \ge \sup_{\mathrm{Bx}(Z)} \partial_h b^-_r$$

be interval enclosures of the derivative $\partial_h(B/A) = (A\,\partial_h B - B\,\partial_h A)/A^2$. Since the vertical segment $\{(p,q)\} \times [0,h]$ lies in $\mathrm{Bx}(Z)$ for every point of Z, the mean value theorem gives

$$b^+_r(h) \ge b^+_r(0) + h\,\alpha_r, \qquad b^-_r(h) \le b^-_r(0) + h\,\beta_r.$$

This is the Lipschitz condition, made explicit and uniform on the cell.

In words: in the limit $h \to 0$ every chord is an exact affine function of the face coordinates, and for $h > 0$ each endpoint of a chord moves at most linearly in h, at a rate bounded rigorously on the whole cell. The next lemma turns these two facts into a lower bound for the chord that is concave, so that it only has to be checked at the vertices of the cell.

Lemma 13. *Let* $\mathrm{gap}^+_r = b^+_r(0) - b^+_{r_*}(0) \ge 0$ *and* $\mathrm{gap}^-_r = b^-_{r_*}(0) - b^-_r(0) \ge 0$, *and*

$$\varphi = \min_r(\mathrm{gap}^+_r + h\,\alpha_r) + \min_r(\mathrm{gap}^-_r - h\,\beta_r).$$

Then φ *is concave in* (p,q,h), *and the chord* $\ell(h) = \max\big(0, f(h)\big)$, *where* $f(h) = \min \mathcal{B}^+ - \max \mathcal{B}^-$, *satisfies* $\ell(h) - \ell(0) \ge \max(\mathrm{opt}_0, \mathrm{opt}_1)$ *with*

- $\mathrm{opt}_0 = \varphi$ *and* $\mathrm{opt}_1 = -f(0)$ *if* $f(0) \ge 0$ *on* Π;
- $\mathrm{opt}_0 = f(0) + \varphi$ *and* $\mathrm{opt}_1 = 0$ *if* $f(0) \le 0$ *on* Π.

Every convex combination $\kappa\,\mathrm{opt}_1 + (1-\kappa)\,\mathrm{opt}_0$ *with* $\kappa \in [0,1]$ *is concave and is also a lower bound for* $\ell(h) - \ell(0)$.

Proof. φ is a sum of minima of affine functions, hence concave. By the mean value bounds, $\min \mathcal{B}^+(h) \ge b^+_{r_*}(0) + \min_r(\mathrm{gap}^+_r + h\alpha_r)$ and similarly for $\max \mathcal{B}^-(h)$, so $f(h) \ge f(0) + \varphi$. If $f(0) \ge 0$, then $\ell(0) = f(0)$, so $\ell(h) - \ell(0) \ge f(h) - f(0) \ge \varphi$ and $\ell(h) - \ell(0) \ge -\ell(0) = -f(0)$. If $f(0) \le 0$, then $\ell(0) = 0$, and $\ell(h) \ge f(h) \ge f(0) + \varphi$ and $\ell(h) \ge 0$. A convex combination of

two lower bounds is at most their maximum, and a convex combination of concave functions is concave. □

Proposition 14 (cell certificate)**.** *On a cell Z, let $G(p, q)$ be the limit value of $\mu_{11}(P) - 1$ as $h \to 0$, which is exactly affine on Π, except that for the depth-7/2 tiles the Lebesgue part is replaced by its affine McCormick lower bound described below, let $\kappa_i \in [0, 1]$ be rational numbers, one per segment i of weight w_i, and let*

$$T = G + \sum_i w_i\big(\kappa_i \operatorname{opt}_{1,i} + (1 - \kappa_i) \operatorname{opt}_{0,i}\big) + h\,\alpha_U,$$

where α_U is a lower bound for the h-derivative of the Lebesgue term on $\mathrm{Bx}(Z)$. Then $\mu_{11}(P)-1 \geq T$ on Z, and T is concave. Hence if $T \geq 0$ at the vertices of Z, evaluated in exact rational arithmetic, then $\mu_{11}(P) \geq 1$ for every pose P in Z.

Proof. The mass is the sum of the segment contributions $w_i\ell_i$ and the Lebesgue term, so the first claim follows from Lemma 13 and the derivative bound for the Lebesgue term described below. A concave function on a polytope attains its minimum at a vertex. □

The coefficients κ_i are proposed by a floating-point linear program and then used as exact rationals; the linear program plays no role in soundness. Only the depth-7/2 tiles reach U_{11}. For them, the limit value of the area $|P \cap U_{11}|$ is a product of two clamped affine functions; it is bounded below by its McCormick envelope after splitting Π at the kinks. Its h-derivative is the integral of the normal velocity of ∂P over $\partial P \cap U_{11}$; each edge of P is clipped against U_{11} like a segment, and the integral is enclosed by intervals.

A cell that fails is split, by bisection or along a *kink plane* $h = -f(0)/\dot{f}$, where a segment's bound changes branch ($\dot{f}$ being the slope in h of its bound), down to a side length of 2^{-30}. A cell still refused at that size is reported as a failure, never discarded.

6.6. **Corner points.** At isolated points of a face two events coincide, for instance two edges of the tilted square crossing segment endpoints at the same time; there the bound fails at every scale on the cells that touch the point. These points have rational face coordinates (p^*, q^*) with small denominators. The checker groups the refused cells by such a point, takes the smallest dyadic ρ for which the box $[p^* - \rho, p^* + \rho] \times [q^* - \rho, q^* + \rho]$ contains them all, and re-covers that box by a second blow-up, again with scale h. Write $X = |p - p^*|$, $Y = |q - q^*|$ and $M = 4$.

zone	poses	face coordinates	slab
center	$X, Y \leq Mh$	$(\bar{p}, \bar{q}) \in [-M, M]^2$, $(p, q) = (p^*, q^*) + h(\bar{p}, \bar{q})$	$h \leq u_0$
X-far	$X > Mh \geq Y$	$(X, \bar{q})$, $q = q^* + h\bar{q}$	$h \leq X/M$
Y-far	$Y > Mh \geq X$	$(\bar{p}, Y)$, $p = p^* + h\bar{p}$	$h \leq Y/M$
both far	$X, Y > Mh$	(X, Y), two triangles	$h \leq \min(X, Y)/M$

(each far zone for both signs of $p - p^*$ and $q - q^*$). In a wedge chart, with slab $h \leq q/K$, the corner points lie on the edge $q = 0$, the box is $[p^* - \rho, p^* + \rho] \times [0, \rho]$, and there are three zones: center ($X \leq Mh$, face $(\bar{p}, q)$ with $p = p^* + h\bar{p}$, slab $h \leq q/K$); far, above the diagonal ($X > Mh$, $q \geq X$, slab $h \leq X/M$); and far, below the diagonal ($X > Mh$, $q \leq X$, slab $h \leq q/K$). Where it was needed (two corners in each of two jobs), the last zone is covered once more by five zones around a ray $q = \lambda X$ through the corner point ($\lambda = 1/2$): with $\Delta = q - \lambda X$ and $M' = 1$, a near zone $|\Delta| \leq M'h$ (face $(X, (q - \lambda X)/h)$, slab $h \leq \lambda X/(K - M')$), and on each side of the ray a zone where $|\Delta| > M'h$ with slab $h \leq |\Delta|/M'$ and a zone where the original slab $h \leq q/K$ is kept, split where $|\Delta|/M' = q/K$.

Lemma 15 (the corner zones cover)**.** *Each family of zones covers the set of poses it replaces: the box times the slab of the original chart. (For the first family, $\rho \le Mu_0$ is required, and checked by the program.)*

Proof. For the first family, every pose satisfies exactly one of $X \le Mh$ or $X > Mh$, and one of $Y \le Mh$ or $Y > Mh$. In the center zone $(\bar{p}, \bar{q}) = ((p - p^*)/h, (q - q^*)/h) \in [-M, M]^2$ and $h \le u_0$ is the original slab. In the far zones the defining strict inequalities give the slab bounds, which are at most $\rho/M \le u_0$. For the wedge corners, a pose with $X \le Mh$ is in the center zone with $\bar{p} \in [-M, M]$ and the original slab; a pose with $X > Mh$ and $q \ge X$ has $h < X/M$; a pose with $X > Mh$ and $q \le X$ keeps $h \le q/K$. For the ray zones, take a pose with $h \le q/K$. If $|\Delta| \le M'h$, then $q \le \lambda X + M'h$, and $Kh \le q$ gives $h \le \lambda X/(K - M')$. If $\Delta > M'h$, then $h < \Delta/M'$, and the pose lies in the zone with that slab when $\Delta/M' \le q/K$, and in the zone that keeps $h \le q/K$ otherwise. The case $-\Delta > M'h$ is the same. Some zones contain more poses than needed; that is harmless, because the pose formulas are polynomial identities and Proposition 14 is proved on each zone as a whole. □

All zones are again charts in the sense above: the center of the pose is polynomial in the face coordinates and h, so they are certified by Proposition 14.

6.7. **Computation.** The poses of Theorem 11 are split into 308 jobs at $u_0 = 1/4000$, one per base line or strip, depth and tilt sign, and 15 more at $u_0 = 1/2000$. Each job reports every chart and every corner zone. In total about 2.75 million cells were certified. Two jobs, at depth $7/2$ and $y_b = 3.5$, needed the wedge-corner and ray zones; one job at $u_0 = 1/2000$ needed four corner boxes. Every chart and every corner zone of every job is certified, with no refused cell. This proves Theorem 11.

6.8. **Coverage of the remaining boxes.**

Proposition 16. *Every sub-box that* `zmx2` *leaves uncertified in the* 75 *tight-tile root boxes consists of poses covered by Theorem 11 or of untilted poses.*

Proof. `zmx2` was rerun on these root boxes with the same settings, printing all 1,255,023 uncertified sub-boxes, which have exact rational ends in center and u. A sub-box near $x = d_0$ consists of left-wall tile poses with $\sigma = +1$. A sub-box near $y = d_0$ is mapped to left-wall poses by the reflection $(x, y) \mapsto (y, x)$, which preserves μ_{11} and reverses orientation, so $\sigma = -1$ and the height is the original x. A script checks in exact arithmetic that every sub-box satisfies $u \le u_0$, has offset at most $\delta_{\max}$ from d_0, and lies in the certified height ranges (for $u \le 1/2000$ also the five further ranges with their extensions); its points with $u = 0$ are untilted. The same script re-reads every job output and requires each expected chart to be present, run with the stated parameters and free of refused cells. All 1,255,023 sub-boxes pass, and all 308 jobs at $u_0 = 1/4000$ and all 15 at $u_0 = 1/2000$ pass the job check. □

7. Controls and independent checks

The controls below are not part of the proof. They test the checkers: a checker that never refuses proves nothing. Each control runs a deliberately *modified* measure, not μ_{11}, through a checker on a region where μ_{11} itself is certified, and the checker must refuse it.

- `zmx2`, at depth 24 on the root box $(15, 36)$ of the final measure, certifies μ_{11} (0 uncertified sub-boxes) and refuses the measures with every segment weight multiplied by 0.999 (209 uncertified) and by 0.993 (217).

- `qx2_zm.py` on the root box of part (3) of Theorem 10, with every segment weight multiplied by 0.999, leaves 87 sub-boxes uncertified (against 83 for μ_{11}), and three of them meet sub-boxes that `zmx2` leaves; so the closure argument of part (3) fails for the modified measure, as it must.
- The tight-tile checker refuses three modified measures on regions where μ_{11} is certified: all weights multiplied by 0.99; the segments on the line $x = 2$ weakened by 1%; and the subtlest one, a measure with the *same* limit values as μ_{11} but a smaller first-order growth. It is built from μ_{11} by moving one segment ($x = 1.9$, $y \in [5.0, 5.1]$, weight 0.0142), which touches the tile centered at $(3/2, 9/2)$ only at a point, to $y \in [5.1, 5.2]$. This changes no limit value at that tile, but the modified measure is no longer valid: the pose with center $(3/2+h,\ 9/2+h)$ and $u = h$ has exact mass $1 - 1.34 \cdot 10^{-7}$ at $h = 10^{-6}$, while under μ_{11} the same pose has mass $1 + 1.22 \cdot 10^{-7}$. A checker that looked only at limit values could not tell the two measures apart. On the same region of the face, the tight-tile checker certifies μ_{11} with no refused cell (155 cells processed), and refuses 1,190 of the 6,079 cells it processes for the modified measure, including those containing the violating poses. So it detects first-order deficits, as the proof requires.
- An independent review of the tight-tile proof and code recomputed $\mu_{11}(P) - 1$ in exact rational arithmetic, with all 13,244 segments and the exact area of $P \cap U_{11}$, at every vertex of every certified cell in sample regions, including all corner and ray zones, and found $T \leq \mu_{11}(P) - 1$ throughout. It also tested the interval arithmetic at 44,000 adversarial points and sampled the covering case analyses exactly, with no failures.
- A floating-point evaluation of μ_{11}, written from scratch, at 90,000 random, 90,000 near-tile and 90,000 near-45° poses found no mass below 1 beyond rounding ($1 - 9 \cdot 10^{-15}$, in the Lebesgue middle, where the mass is exactly 1).

8. How the measure was found

This section is not part of the proof.

We minimized $\mu_k([0, k]^2)$ over the family by Daniel's quadrant linear program `qx2_lp.py` [3]: segment variables on the 1/10 grid with $R = 4$, exact rows for the limits $\theta \to 0$, and a tilt margin (mass at least $1 + 0.05 \min(\theta, \pi/180)\,(1 - |P \cap U_k|)$, θ in radians), so that the checkers have room away from tight poses. Each round projects the linear-programming solution to an exact rational point on its tight rows (by QR pivoting and exact solution with FLINT), runs the exact checks and `zmx2`, and turns uncertified poses into new rows. This oracle loop mattered: random sampling of poses missed real deficits. In one iterate, a root box that `zmx2` refused turned out to contain a pose of exact mass 0.99998777 at a tilt of 0.89°; its neighborhood became new rows. For the final iterate the exact D differs from the floating-point optimum by less than 10^{-7}. Five oracle rounds were run at the final tilt margin; the exact projection of the last one satisfies all 37,451 of its tight rows exactly.

9. Remarks

Remark 17 (why a Lebesgue interior). Far from the walls, a measure made only of points and segments that gives every unit square mass at least 1 must spend more than mass 1 per unit area. Lebesgue measure gives every unit square mass exactly 1, so with a Lebesgue interior the middle costs exactly its area and the whole saving comes from the corners.

Remark 18 (small k). By linear-programming duality, the least total mass of a valid measure is at least the fractional packing number of $[0, k]^2$ (the largest total weight of a nonnegative weighting of poses giving every point weight at most 1). A cover can therefore prove $s(n) \geq k$

only when that number is below n. For $k = 5, 6$ our floating-point linear programs of this kind stay above $k^2 - 5$, so the theorem starts at $k = 7$.

Remark 19 (beyond five). Proving $s(k^2-6) = k$ in the same way would need $D > 3/2$; at $R = 4$ the linear program stays near 1.34, and larger corners may help. The Tight-Tile Lemma is not specific to squares: any weighted cover of a container with straight walls has tight wall tiles, and the same blow-up in the tilt applies, for instance to equilateral triangles in a triangle, with arithmetic in $\mathbb{Q}(\sqrt{3})$.

Acknowledgments and AI assistance

The search, the certificate, the Tight-Tile checker and its proof, and the verification runs were produced with Claude (Anthropic) under the authors' direction; the authors reviewed the method, the checks and the text. We thank Evan Daniel for the open-source linear program, checkers, format and Lean reductions this work builds on; Hiroaki Hosono and B. Greff for their public certificates; and the Squares Project for its records.

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